CSCE 3701 · TEAM CLOSET · MILESTONE 2 · SPRINT 1

# *Closet*

## Test Plan and Report · Version 2.0

Build v1.6 + Virtual Try-On (US-118) · November 4, 2026

# 1. Test plan

**Objective.** Show that every requirement in the SRS v2.0 works as specified, that nothing that worked before broke, and that the new Virtual Try-On is safe (another user cannot reach my photos) and fails gracefully.

| Level | What | Tool | Where it runs |
|---|---|---|---|
| **Unit** | Pure rules: validators, recommendation engine, lifecycle state machine, Ask Closet parser, Virtual Try-On service and providers | Jest 29 | every push (GitHub Actions) |
| **API / integration** | The real Express app through the security pipeline against a real PostgreSQL 16 database (emptied before each test) | Jest + Supertest | GitHub Actions (PostgreSQL service) |
| **Acceptance** | One test per Sprint 1 story, written from its acceptance criteria | Jest + Supertest | GitHub Actions |
| **End-to-end (browser)** | Real user journeys clicked in Chromium: demo, stateful, lifecycle, virtual try-on | Playwright (Python) | GitHub Actions + offline demo |
| **Manual** | Demo rehearsal on the presenting laptop; visual check of screens | — | team |

**Techniques.** Equivalence partitioning and boundary values (photo size and type, 1 – 6 pieces, daily limit), state-transition testing (81 ordered pairs of the 9 lifecycle states; PENDING → PROCESSING → COMPLETED / FAILED), decision tables (deletion reasons), scenario tests (the black blazer at the tailor), and security cases (no token → 401, someone else's data → 404). **Test doubles:** the weather, Google sign-in and the AI try-on provider are replaced by fakes, so tests need no keys and no network. **Entry / exit:** a story is Done when its acceptance test and the full suite pass on GitHub Actions.

# 2. Results

| Suite | Result | Run |
|---|---|---|
| **Full suite, build v1.6 (unit + API + acceptance)** | **599 / 599 passed** · 0 failed · line coverage 97.53 % | GitHub Actions CI #5, 29 Oct 2026 (real Jest, PostgreSQL 16) |

| Suite | Result | Run |
|---|---|---|
| **Virtual Try-On unit tests VT-01 – VT-27** | **23 / 23 passed** | sandbox, 4 Nov 2026 (Node 22; Jest-compatible runner because the npm registry is blocked there) |
| **Full suite after Virtual Try-On (GitHub Actions, 4 Nov)** | 630 / 631 passed · 1 failed: OF-09 (defect D-33) · line coverage 94.53 % | GitHub Actions run 41865197065, real Jest + PostgreSQL 16 |
| **After the fixes (GitHub Actions CI #20, 5 Nov)** | **all four jobs green: server (631 tests), client, e2e, offline demo** | GitHub Actions CI #20, commit 2e8ab78 |
| **Virtual Try-On API tests API-VT01 – VT09** | **9 / 9 passed** | GitHub Actions run 41865197065 |
| **e2e virtual_tryon_journey.py** | virtual try-on journey passed on the offline demo: 13 steps | offline demo build, Chromium, 4 Nov 2026 |
| **e2e lifecycle_journey.py (regression)** | offline demo: 11 checkpoints, 0 browser errors | offline demo build, 4 Nov 2026 |
| **e2e stateful_journey.py (regression)** | offline demo: 14 checkpoints, 0 browser errors | offline demo build, 4 Nov 2026 |

**Honest status.** The old try-on tests TO-01 – TO-12 were removed with the old `/tryon` route they tested; the new Virtual Try-On replaces both. The real FASHN service has not been called yet (no API key in the sandbox): the provider was replaced by a test double and a mock. The live provider test is task T-30.

## 2.1 Virtual Try-On · unit tests (passed)

| Test | What it shows |
|---|---|
| **VT-01** | FASHN: submits the person photo + garment with the category and returns the job id |
| **VT-02** | FASHN: status maps processing → PROCESSING, completed → COMPLETED + result URL, failed → FAILED |
| **VT-03** | FASHN: no key → disabled; bad key, rate limit and network errors become typed ProviderErrors |
| **VT-04** | provider choice comes from the environment; mock only when asked for; every provider has the full contract |
| **VT-05** | real type and pixel size are read from the file header; non-images are refused |
| **VT-10** | successful try-on: PENDING → PROCESSING → COMPLETED, result stored privately, original garment photo untouched |
| **VT-11** | an async provider is polled through GET: still PROCESSING until the job finishes |
| **VT-12** | invalid or someone else's wardrobe item → 400 / 404 (nothing is sent) |
| **VT-13** | unsupported garment type (shoes) → 422 UNSUPPORTED_GARMENT |

| Test | What it shows |
|---|---|
| VT-14 | missing images: no user photo → PHOTO_REQUIRED; piece without a usable photo → GARMENT_IMAGE_MISSING / UNSUITABLE |
| VT-15 | user photo validation: not an image → 415, too small → 422, no consent → 400 |
| VT-16 | provider failure → the try-on is FAILED with a friendly message, and Retry starts it again |
| VT-17 | timeout, failed job and empty result all end as FAILED (never stuck, never blank) |
| VT-18 | a temporary network error while polling keeps it PROCESSING (the next poll tries again) |
| VT-19 | status retrieval and isolation: another user can neither read, save, retry nor delete my try-on |
| VT-20 | save result, list saved, delete removes the stored images |
| VT-21 | saved try-on photo: save, reuse, replace, delete |
| VT-22 | uploading with "save this photo" also stores it for next time |
| VT-23 | limits: daily quota and at most two running at once → 429 |
| VT-24 | provider not configured → 503 TRYON_DISABLED and status says enabled: false |
| VT-25 | a saved look links the try-on to the outfit; someone else's look is 404 |
| VT-26 | an https result from the provider is copied into private storage |
| VT-27 | account deletion removes every try-on file of the user |

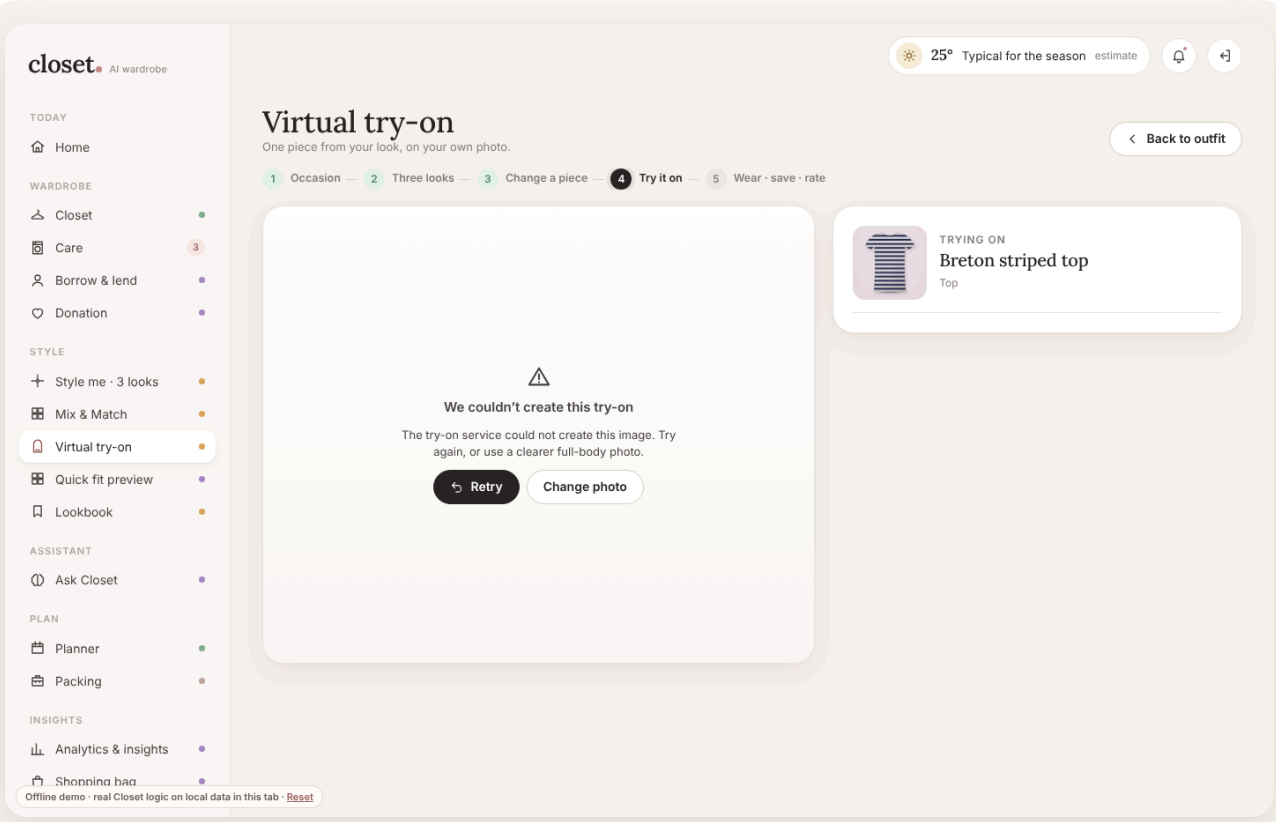
## 2.2 Virtual Try-On · API tests (run in CI)

| Test | What it shows |
|---|---|
| API-VT01 | without a login token every route is 401 |
| API-VT02 | create → 202 PROCESSING, then GET → COMPLETED with a signed result URL that serves an image |
| API-VT03 | invalid item id → 400; unknown item → 404; shoes → 422 UNSUPPORTED_GARMENT |
| API-VT04 | missing photo → 400 PHOTO_REQUIRED; a non-image → 415; no consent → 400 |
| API-VT05 | provider failure → FAILED with a friendly error, retry works once the provider is back |
| API-VT06 | another user cannot read, save, retry or delete my try-on (404) |
| API-VT07 | save the result, list saved, delete it |
| API-VT08 | saved try-on photo: PUT, reuse with useSavedPhoto, DELETE |
| API-VT09 | status route; with no provider configured, creating is 503 |

## 2.3 Virtual Try-On · browser journey

| Step | Checked |
|---|---|
| v01 | Every one of the 3 looks shows Virtual Try-On, Wear this, Save, Not for me and Preview |
| v02 | The picker lists the look's pieces; only tops, jackets and dresses are enabled |
| v03 – v04 | No photo → message; too-small photo and non-image rejected in the browser |

| Step | Checked |
|---|---|
| **v05** | Generate without consent → message; consent + save photo |
| **v06 – v07** | Loading state, then the result with its label; Save result stored (API checked) |
| **v08** | Try another item offers the saved photo first |
| **v09 – v10** | Forced provider failure → friendly error → Retry → success |
| **v11** | Leaving while it runs; “Your try-ons” lists it and it completes |
| **v12** | Back to outfit returns to the same three looks |
| **v13** | Delete saved photo (API checked) |




*Error state with Retry (step v09).*

## 2.4 Defects found in this pass (all fixed)

| ID | Found by | Severity | Description | Fix |
|---|---|---|---|---|
| **D-29** | e2e v05 | High | Offline demo: deleting a seeded piece removed a photo shared by other pieces, so a top had “no usable photo”. | only photos added in the tab are removed |
| **D-30** | visual check | Medium | The garment picker opened inside the flipping outfit card and was clipped. | rendered into the page body (portal) |
| **D-31** | e2e v08 | Low | Try another item on the same piece kept the previous upload instead of the saved photo. | reset photo choice and consent |
| **D-32** | e2e v12 | Medium | Back to outfit from “Your try-ons” opened an empty Style me. | returns to the stored three looks |

| ID | Found by | Severity | Description | Fix |
|---|---|---|---|---|
| D-33 | CI (OF-09) | Low | The seeded presentation plan had a fixed 24 °C; in November the seasonal estimate is 25 °C, so the “warmer forecast” alert no longer appeared. | plan = seasonal estimate − 6 °C, every month |
| D-34 | CI (demo check) | Low | The demo check expected a seasonal-storage suggestion, which is only made in summer or winter. | expected only in summer / winter |

## 2.5 Not verified yet

- Live FASHN generation (needs the team’s API key, T-30).
- Usability test with classmates (T-33) and load test (T-34) — Sprint 2.